

Appendix D - Beagle Marine Park Data Analysis
Report (stereo BRUV)
South-east Marine Parks Network

Table of contents

Data Analysis	1
Benthic ecosystem extent	2
Demersal fish assemblage	6
Significance of year and zone status	11

Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A.

Benthic ecosystem extent

Table D.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc.

Response	Model	ΔAICc	ωAICc	R^2	EDF
Sand	year + geoscience_roughness + geoscience_aspect(year) + geoscience_detrended(year)	0	0.423	0.05575	10.45
	year + geoscience_aspect(year) + geoscience_detrended(year) + geoscience_roughness(year)	0.878	0.273	0.05641	11.75
	year + geoscience_detrended + geoscience_aspect(year) + geoscience_roughness(year)	1.238	0.228	0.05511	10.19
Sessile invertebrates	year + geoscience_roughness + geoscience_aspect(year) + geoscience_detrended(year)	0	0.347	0.05483	10.44
	year + geoscience_detrended + geoscience_aspect(year) + geoscience_roughness(year)	0.392	0.285	0.05453	10.24
	year + geoscience_aspect(year) + geoscience_detrended(year) + geoscience_roughness(year)	0.656	0.25	0.05564	11.79
Reef	year + geoscience_roughness + geoscience_aspect(year) + geoscience_detrended(year)	0	0.461	0.05675	10.44
	year + geoscience_aspect(year) + geoscience_detrended(year) + geoscience_roughness(year)	0.897	0.294	0.05747	11.79
	year + geoscience_detrended + geoscience_aspect(year) + geoscience_roughness(year)	1.802	0.187	0.05593	10.24

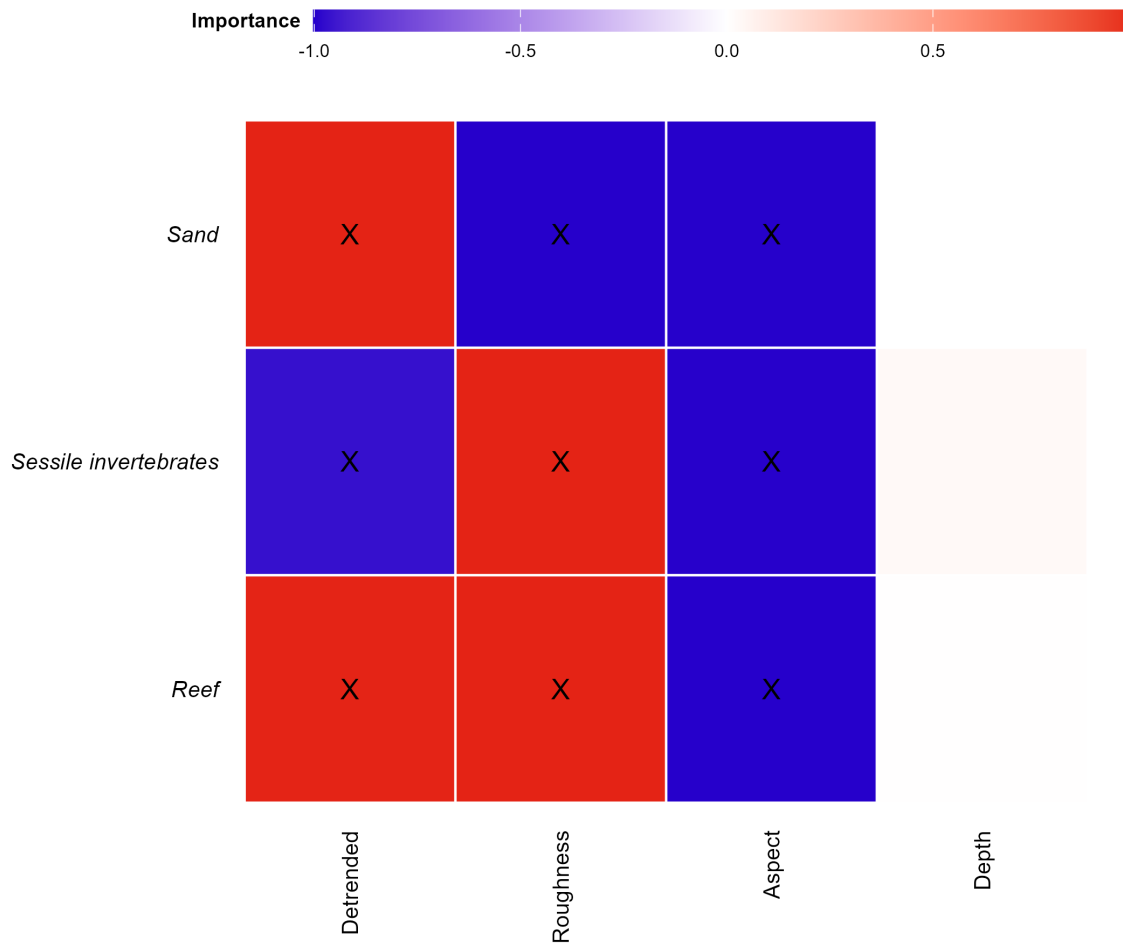


Figure D.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

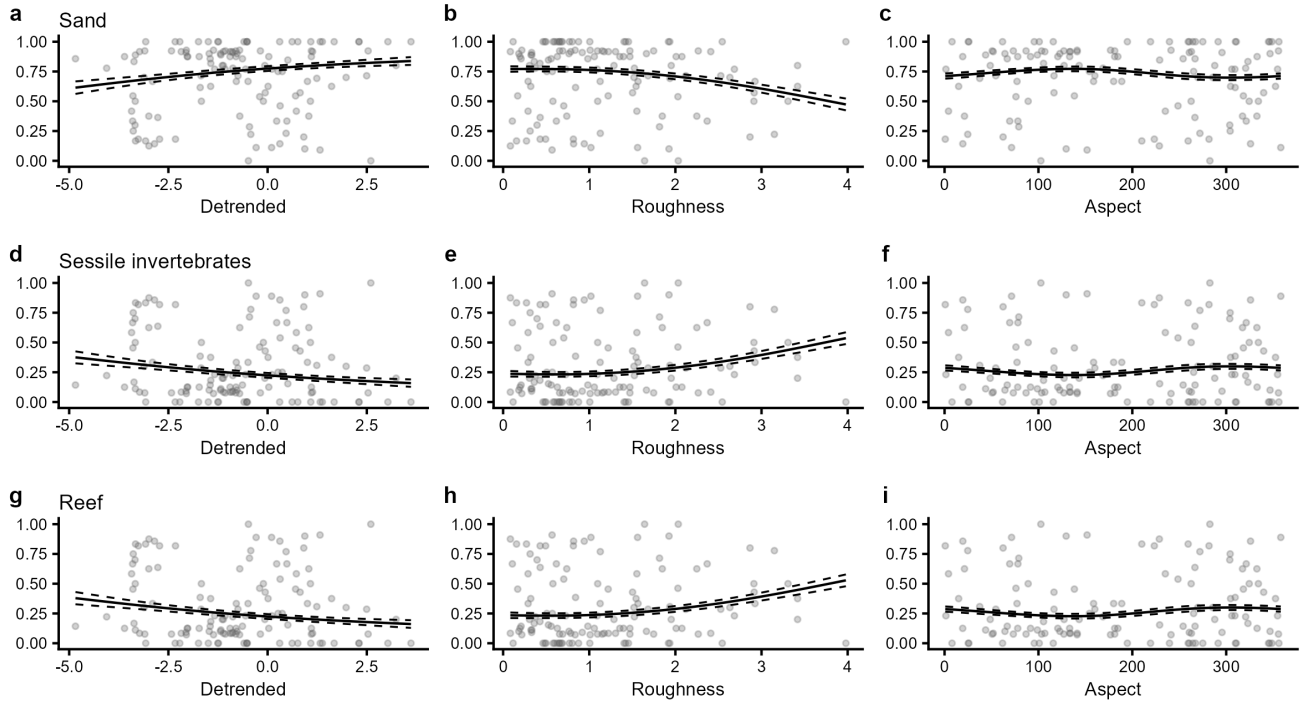


Figure D.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years.

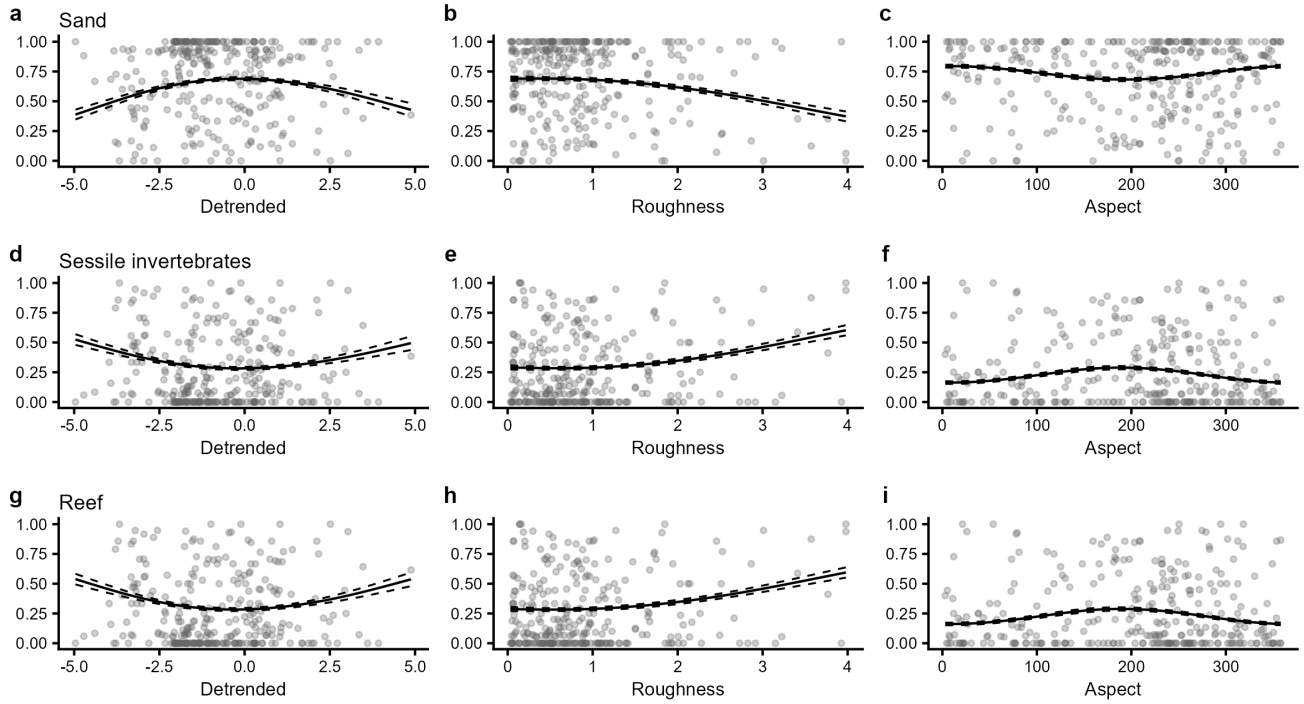


Figure D.3: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years.

Demersal fish assemblage

Table D.2: Demersal fish model selection results. Top generalised additive models (GAMs) for predicting the species and abundance distribution of metrics of interest from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the best model (fewest variables - shown in bold) within two units of the lowest AICc. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Response	Model	ΔAICc	ωAICc	AICc	R^2	EDF
Species richness	detrended + aspect(year) + Z(year) + reef + year + status	0	0.141	2195.32	0.28899	9.91
	detrended + aspect(year) + Z(year) + reef(year) + year + status	0.163	0.13	2195.48	0.29303	11.20
	detrended(year) + aspect(year) + Z(year) + reef + year + status	1.189	0.078	2196.51	0.29024	10.86
	detrended + Z(year) + reef + year + status	1.19	0.078	2196.51	0.28062	7.57
	detrended(year) + aspect(year) + Z(year) + reef(year) + year + status	1.215	0.077	2196.53	0.29436	12.12
	detrended + aspect + Z(year) + reef + year + status	1.754	0.059	2197.07	0.28262	8.57
	detrended + Z(year) + reef(year) + year + status	1.901	0.054	2197.22	0.28374	8.82
Total abundance	detrended(year) + Z(year) + reef(year) + year + status	0	0.115	3602.62	0.43434	11.31
	detrended(year) + aspect(year) + Z(year) + reef(year) + year + status	0.005	0.114	3602.62	0.43435	13.31
	detrended(year) + aspect + Z(year) + reef(year) + year + status	0.039	0.112	3602.66	0.43439	12.31
	Z(year) + reef(year) + year + status	0.423	0.093	3603.04	0.42631	8.65
	aspect(year) + Z(year) + reef(year) + year + status	0.425	0.093	3603.04	0.42631	10.65
	aspect + Z(year) + reef(year) + year + status	0.442	0.092	3603.06	0.42633	9.65
	roughness(year) + Z(year) + reef(year) + year + status	1.379	0.057	3604.00	0.43485	11.79

(continued)

Response	Model	Δ AICc	ω AICc	AICc	R^2	EDF
	roughness(year) + aspect(year) + Z(year) + reef(year) + year + status	1.393	0.057	3604.01	0.43486	13.79
	roughness(year) + aspect + Z(year) + reef(year) + year + status	1.399	0.057	3604.02	0.43487	12.79
	roughness(year) + aspect + Z(year) + reef + year + status	0	0.182	3266.53	0.35509	10.94
	roughness + aspect + Z(year) + reef(year) + year + status	0.422	0.148	3266.96	0.35748	11.52
Large Reef Fish Index*	detrended(year) + aspect + Z(year) + reef(year) + year + status	0.919	0.115	3267.45	0.36404	13.06
	roughness(year) + aspect + Z(year) + reef(year) + year + status	0.944	0.114	3267.48	0.3615	12.39
	roughness + aspect + Z(year) + reef + year + status	1.052	0.108	3267.59	0.35157	10.54
	detrended(year) + aspect + Z(year) + reef + year + status	1.95	0.069	3268.48	0.3571	12.17
Reef Fish Thermal Index	aspect(year) + Z(year) + reef(year) + year + status	0	0.16	850.07	0.2557	10.70
	aspect + Z(year) + reef(year) + year + status	0.056	0.156	850.13	0.25586	9.77

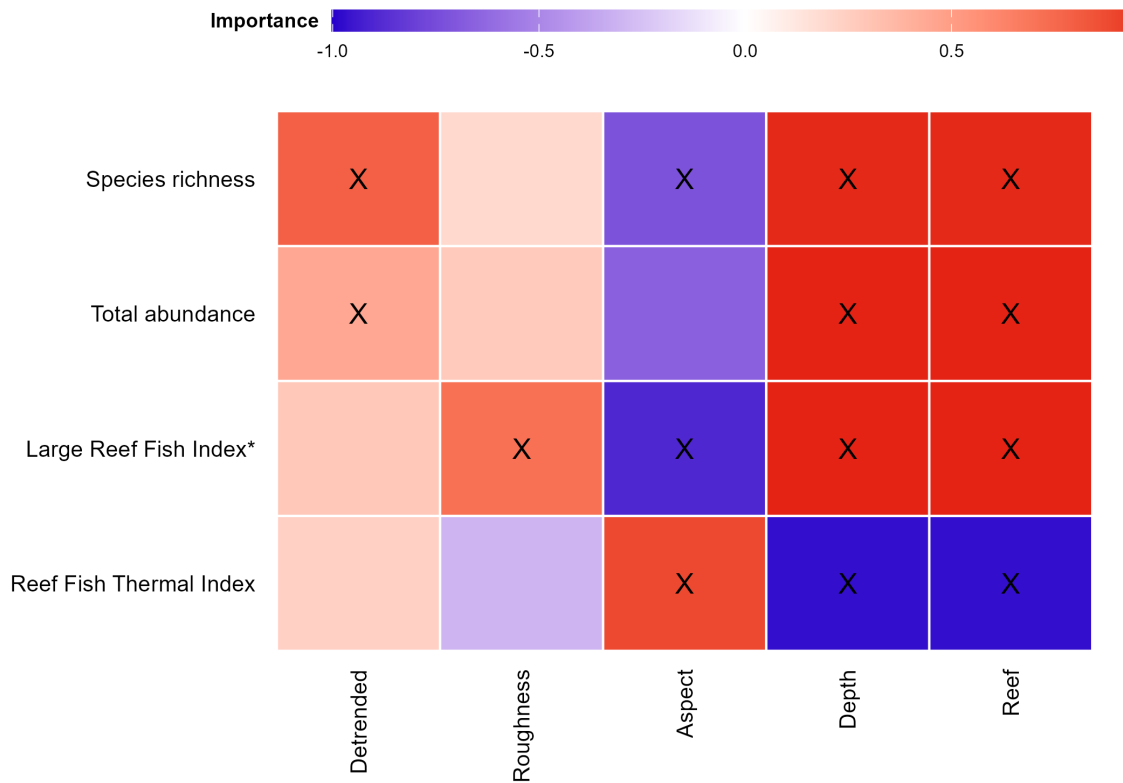


Figure D.4: Variable importance scores for predicting the probability of demersal fish metrics. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

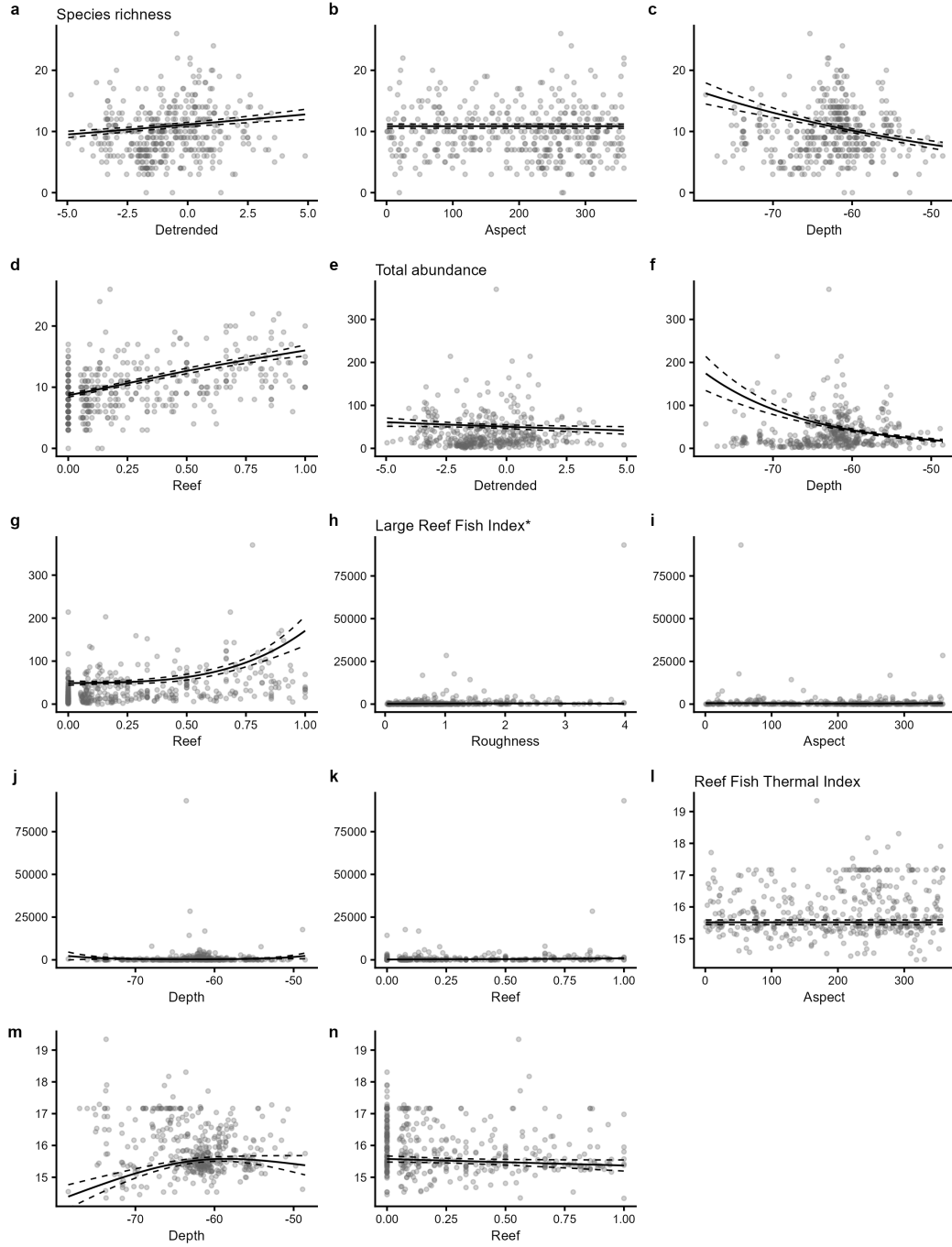


Figure D.5: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with year held at 2018 and zone status held at areas open to fishing. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

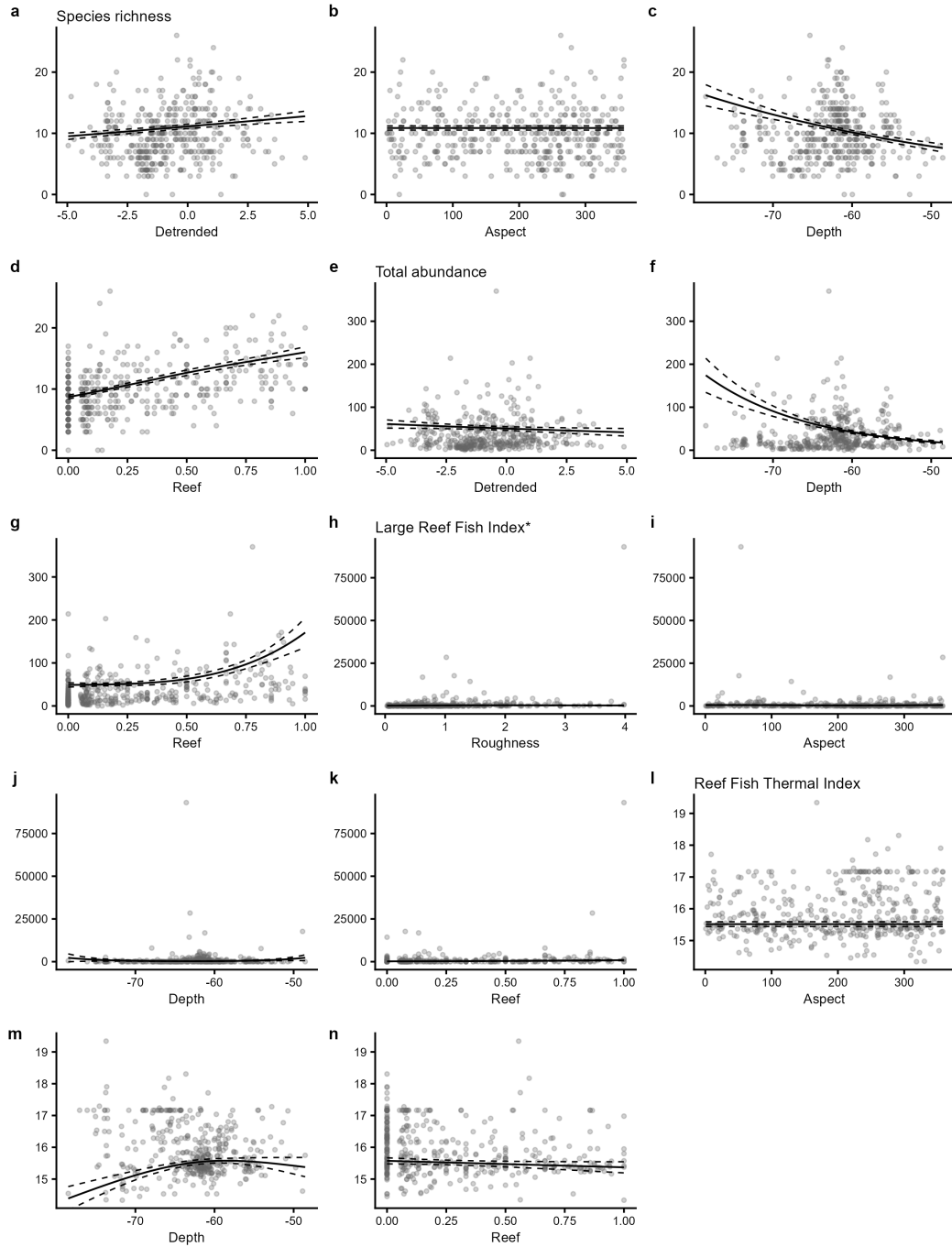


Figure D.6: Plots of the best model found to predict the probability of occurrence of key fish groups. Individual panes display the predicted relationships between model covariates and each demersal fish metric. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value, with year held at 2018 and zone status held at areas open to fishing. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data from all survey years and zones. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

Significance of year and zone status

Table D.3: Significance of the year and zone status terms in each demersal fish GAMM. Estimate, standard error (SE), test statistic and p-value are taken from the parametric coefficient table of each model summary, one row per factor-level contrast (e.g. a survey year against the reference year, or No-Take against Fished).

Response	Term	Estimate	SE	Statistic	p-value
Species richness	Zone status: No-Take vs Fished	-0.108	0.095	-1.135	0.256
	Year: 2025 vs 2018	-0.119	0.035	-3.414	<0.001
Total abundance	Zone status: No-Take vs Fished	0.232	0.198	1.171	0.242
	Year: 2025 vs 2018	-0.886	0.074	-11.906	<0.001
Large Reef Fish Index*	Zone status: No-Take vs Fished	-0.921	0.630	-1.462	0.145
	Year: 2025 vs 2018	0.338	0.242	1.396	0.164
Reef Fish Thermal Index	Zone status: No-Take vs Fished	0.158	0.192	0.823	0.411
	Year: 2025 vs 2018	0.592	0.078	7.596	<0.001

Table D.4: Zone status test for each demersal fish metric, laid out by survey year. Zone status (No-Take vs Fished) is fitted as a single term across all years (no year by status interaction), so the status estimate, SE and p-value are the same in every row for a given metric; predicted values and their SE are shown per year, with other model covariates held at their mean.

Response	Year	Fished (predicted)	Fished (SE)	No-Take (predicted)	No-Take (SE)	Status estimate	Status SE	Status p-value
Species richness	2018	10.865	0.345	9.755	0.965	-0.108	0.095	0.256
	2025	9.813	0.340	8.810	0.834	-0.108	0.095	0.256
Total abundance	2018	51.985	4.250	65.535	14.028	0.232	0.198	0.242
	2025	33.213	2.495	41.870	8.326	0.232	0.198	0.242
Large Reef Fish Index*	2018	298.291	85.699	118.813	81.900	-0.921	0.630	0.145
	2025	873.972	163.249	348.114	218.827	-0.921	0.630	0.145
Reef Fish Thermal Index	2018	15.516	0.074	15.674	0.205	0.158	0.192	0.411
	2025	15.845	0.078	16.003	0.191	0.158	0.192	0.411